

Experiment 6: Time Series Analysis of Website Traffic

```
import pandas as pd
import matplotlib.pyplot as plt

data = pd.read_csv(
    'website_traffic.csv',
    parse_dates=['Date']
)

data['Page_Views'] = data['Page_Views'].interpolate()
data['Page_Views'] = data['Page_Views'].bfill()
data['Page_Views'] = data['Page_Views'].ffill()

plt.figure(figsize=(10, 4))
plt.plot(data['Date'], data['Page_Views'])
plt.xlabel('Date')
plt.ylabel('Page Views')
plt.show()
```

Output:

